

Evidentiary reasoning with the Analytical Hierarchy Process: Use of graph entropy to rank decision elements

unpublished working paper

GitHub: hamlett-neil-ur/ahp-graph-entropy

Neil A. Hamlett, D.Sc., MBA



neil.hamlett@uncertainty-research.science



neil-hamlett-strategic-quant



hamlett-neil-ur.github.io



ORCID: 0000-0001-8278-0087

May 19, 2026

Abstract

Decision makers often require insight into the relative strength of evidence influencing their choices. The Analytic Hierarchy Process (AHP) contains innate mechanisms leading to the rigorous quantification of evidentiary elements' relative evidentiary strength. Pairwise-comparison analysis produces lists of edges for a complete, directed, acyclic graph, often called a tournament graph. Intensities of importance from comparative judgements between evidentiary elements provide tournament-graph edge weights, leading to a novel Laplacian matrix. Relative-entropy calculations based on Laplacians for induced subgraph quantify the relative evidentiary strength associated with the decisional evidence omitted from them. A previously published case study related to a fuel-cell design demonstrates the approach. The graph-entropy method provides clearer strength-of-contribution indications than sensitivity analysis from the case study.

Introduction.

Decision-making nearly always occurs under conditions of incomplete or imperfect information. No less than Nobel-prize economist Frederick von Hayek recognized this:

...I prefer true but imperfect knowledge, even if it leaves much undetermined and unpredictable, to a pretence of exact knowledge that is likely to be false (emphasis added) (von Hayek (1974), cited in Kay & King (2020)).

The treatment of uncertainty in quantitative decision making represents a formidable research challenge con-

fronting multiple facets. We conventionally decompose uncertainty into epistemic and aleatory (or stochastic) components (Salicone & Prioli (2018)).

The present study focuses primarily on the epistemic. Hamlett (2019, 2025) argues for the primacy of resolving epistemic uncertainty in business decision making. Exemplified by Zellweger & Zenger (2023); Rindova & Martins (2024); Ehrig, et al (2025), this perspective holds sway in the entrepreneurial and strategic decision-making communities. Jackson (2019) corroborates from a philosophy-of-science point of view.

Motivation: How strong is our evidence?

Ehrig, et al (2025) argue that entrepreneurial decision making in particular inescapably requires "...logic and reasoning with the aid of very small samples of data" (emphasis added). Circumstances demanding this sort of reasoning arise in many contexts beyond new-business formation (Martin & Olsberg (2007); Eisenmann (2013); Swanson (2024)). Fundamentally, entrepreneurial ventures amount to epistemic explorations. Howard & Abbas (2016) devoted considerable attention to the adverse effects of imperfect information — an endemic condition within entrepreneurship — when trying to select optimum scenarios.

Nontrivial challenges inevitably arise even when considerable amounts of statistically coherent, phenomenologically illuminating data exist. Cover & Thomas (2006) derive an information-theoretic data-processing inequality: "...no processing of Y ... can increase the information that Y contains about X ." Corroborating from a statistical-learning perspective, Hastie, Tibshirani, Friedman (2009) define irreducible variance. Kay & King (2020) warn of pervasive nonstationarity in real-world problems: Statistical-learning training data based

on past experience might not represent future conditions very well. More generally, Townsend, et al (2023) articulate the case for “...computationally irreducible barriers to the predictive capabilities and task performance advantages of AI systems...” for many tasks.

Extending Dempster’s (1967) seminal concept, Shafer (1976) offered a rigorous approach to quantifying epistemic uncertainty. More-recently, Shenoy (2023) applied Shafer’s reasoning to quantify evidentiary incompleteness in Bayesian networks, a causal-inference methodology commonly employed within the entrepreneurial decision-making community (Camuffo, et al (2024); Ehrig, et al (2025)).

Hué, et al (2026) use a stochastic-detection area-under-the-curve (AuC) approach (cf. Fawcett (2026)) to analyze evidentiary factors’ relative strengths. Resembling stepwise regression (Olive (2017)), their approach also exhibits conceptual parallels to Shafer (1976). Cox (2026a,b) offers an emerging approach to uncertainty characterization based on sheaf theory (Rosiak (2022)).

Why the Analytical Hierarchy Process (AHP)?

AHP provides an axiomatic framework (Saaty (1986)) for “...logic and reasoning with the aid of very small samples of data” (Ehrig, et al (2025)). It is a multi-criterion decision methodology (MCDM) enjoying widespread applicability to group decision making (Saaty (1994)), risk analysis (e.g., Saaty (1987); Vaidya & Sushil (2006); Linkov, et al (2006)), and other problems. AHP is often applied to situations where substantial amounts of statistically coherent measurements are not immediately available to support more-empirical approaches.

The AHP methodology begins with identifying a plausible evidentiary span: The range of measurable factors — elements in the vernacular of Saaty (1986, 1994) — believed to influence the attractiveness of each distinct scenario under consideration. A hierarchy model represents this as a tree graph. The responsible individual decision maker or panel of experts then performs comparative judgements: “...the numerical representation of a relationship between two elements that share a common parent” (emphasis added) in the tree graph (Saaty (1994)). A case study in a subsequent section demonstrates AHP’s use to rank plausible scenarios based on such information.

Not strictly a stand-alone method, AHP can be fused with other approaches. Bayesian techniques allow for subsequent revision of the comparative judgements as stronger empirical evidence emerges (Merrick, van Dorp, Singh (2005); Li, H., et al (2025)). Li, J. et al (2025) combined back-propagation neural networks with the

AHP to design shields for neutron generators. Xiao, et al (2025) explored simulation-optimization approaches to circumvent altogether the elicitation of comparative judgements from experts. Lukinskiy, Lukinskiy, Bazhina (2025); Sáenz-Royo & Chiclana (2026) studied methods to improve the robustness of expert inputs into comparative judgements.

Evidentiary-strength analysis using the AHP.

That the comparative judgements for an AHP hierarchy model define a complete, directed, acyclic graph (DAG) provides the seminal insight underlying this investigation. Complete DAGs belong to a special class of graph called tournament graph (Moon (2015)). In general, “the entropy of a graph is interpreted as its structural information content and serves as a complexity measure” (emphasis added) (Dehmer & Mowshowitz (2011); Mowshowitz & Dehmer (2012)). Brown, et al (2020) studied the particular characteristics of tournament graph.

What questions do entropies of AHP-based tournament graph allow us to answer? Endogenous to the AHP formulation itself, relative entropy quantifies the marginal informational contribution by a subset of elements within the evidentiary span. This addresses the objective pursued by Hué, et al (2026). Graph-entropy measures however do not require extensive empirical data spanning of both positive and negative cases, as with Hué’s approach. Congruent with the AHP methodology explored here, Jiroušek, Kratochvíl, Shenoy (2023) use entropy of belief-function graphical models to decompose their elements’ relative evidentiary strength.

By what mechanism do we infer the differential information between hierarchically-coupled AHP hierarchy models? Relative entropy provides a well-understood approach (Cover & Thomas (2006)). Relative entropy for graphs has received considerable attention (Wong & You (1985); Liu, et al (2022); Guanlei, Xiaogang, Xiaotong (2025)). Brown, et al (2020) explored entropy differences between same-order tournament graphs with differing directional structures. In our case, the relative entropy of the graph associated with the overall evidentiary span with respect to a constituent induced subgraph quantifies the information lost due to elements omitted from the latter.

What innovation is required to define a methodology for AHP-based evidentiary-strength analysis? First, we must formulate an entropy definition for tournament graph with weighted edges. Brown, et al (2020) graph with unweighted (uniformly unit-weighted) edges. Chen, et al (2015), Kuo (2022), and Liu, et al (2022) formulate entropies for regular (non-directional) graph comprised of weighted edges. The

intensities of importance from the AHP pairwise comparison provide the weights used here.

Second, relative entropies of hierarchically linked AHP subgraph must be coupled. The *Hierarchy* term in AHP refers to a hierarchic deconstruction of decisions into constituent elements. The AHP method produces a set of decision-element comparative judgements — represented by a tournament graph — for each hierarchy model. (Cover & Thomas, 2006, §2.5) describe chain rules for entropy, relative entropy, and mutual information. Such mechanisms offer a rigorous approach to link hierarchically adjacent tournament graph.

Why is “heavy” graph-entropy machinery needed? We might intuitively expect that individual comparative judgements suffice to inform us about the most-influential elements. The AHP’s machinery however accounts for “mutual coupling” between elements (Saaty (1986)). That comparative judgements define a graph — a network — provides the basis of its power. Our graph-entropy machinery accounts for the aggregate effect of all the evidence informing a decision or risk assessment. Individual importance intensities fail to provide this.

Course of investigation.

We begin with a brief illustration of AHP methodology. A *real-world* case study concisely demonstrates its application to obtaining a preference-ranked set of scenarios. The actual ranking lies beyond our specific interest, but is included for completeness. The DAGs resulting from the comparative judgements provide our point of departure into a novel methodology.

We then calculate the entropies for our comparative-judgement DAGs associated with distinct hierarchy models. This extends Brown, et al (2020) through adoption of a Laplacian structure that includes edge weights. Our AHP-graph Laplacian fuses the structure of a conventional unit-edge-weighted DAG Laplacian with comparative-judgement importance intensities, which provide the edge weights. This incorporates information about degree of influence in the interconnected evidentiary framework.

Finally, we calculate entropies and relative entropies from our AHP-based DAG structure comprised of multiple, hierarchically coupled hierarchy models. We interpret the results for our real-world case study. Approaches analogous to those by Jiroušek, Kratochvíl, Shenoy (2023) guide our thinking. We also explore the relationship between importance intensities and relative entropy. This quantifies relative strengths of contribution by subsets of evidentiary elements.

AHP as path to achieving Decision Quality.

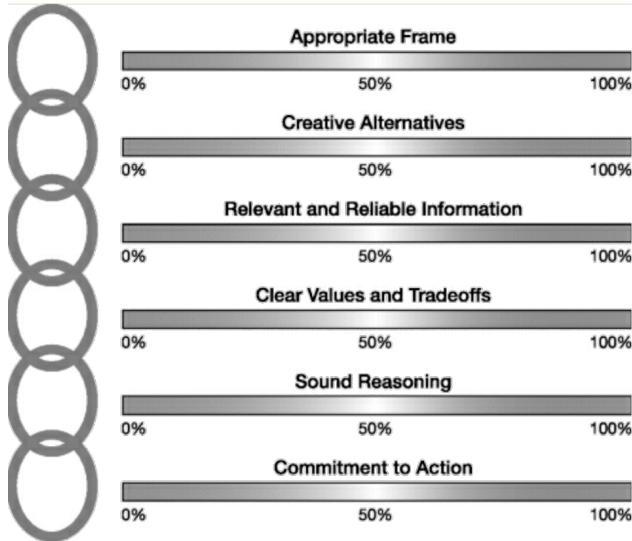
The pattern of reasoning underlying Saaty’s (1986) AHP methodology corresponds to patterns on which prominent applied decision-analysis practices are based. *Decision Quality* (DQ) represents a prominent example (Spetzler, Winter, Meyer (2016)). This context — exogenous to the mechanics of AHP — allow us to highlight our focus on evidentiary reasoning in the context of decision making.

Figure 1 depicts the DQ framework, a *de facto* standards of practice for the Society of Decision Professionals (SDP). SDP is a community of applied decision analysts and scientists. The DQ frameworks contains six links:

- Establishing an *Appropriate Frame* specifies the scope of the decision: What specifically is to be decided. Thoughtful framing combines prescriptive with proscriptive judgements to provide analytical clarity. Saaty (1994) describes this as “...the most creative part of decision making that has a significant effect on the outcome...”
- By definition, the act of deciding involves selection of one or more options from a discrete, precisely defined set of *Creative Alternatives*.
- *Relevant and Reliable Information* constitute the evidence weighed in selecting the best alternative.
- *Clear Values and Tradeoffs* specify the criteria by which the best alternative is identified.
- *Sound Reasoning* defines the logic employed in identifying the best alternative. Creighton (1909) cautioned that “...it is not possible to lay down definite rules for the guidance of thinking in every case.” The reasoning approach must therefore fit the circumstances resulting from the links above. AHP is itself a distinct reasoning approach. ⚠
- *Commitment to Action* represents the follow-through in implementing or adopting the selected alternative. Absent a commitment to adopt or act upon the best alternative resulting from the links above, a decision arguably has not occurred. The analysis was academic.

The DQ framework defines a scientific, evidentiary reasoning pattern that is philosophically consistent with the design-of-experiments practice (Cox & Reid (2000)). Its mention here amplifies the theme of evidentiary reasoning. Saaty (1986, 2003) emphasizes the importance of logical consistency in application of AHP.

Figure 1: Decision-Quality framework. (From Spetzler, Winter, Meyer (2016), Figure 2.1)



Case study in the Analytic Hierarchy Process (AHP).

Bu, et al (2026) employed AHP in making tradeoff decisions during the design of Proton Exchange Membrane Fuel Cells (PEMFCs), which "...efficiently convert chemical energy from fuels into electricity without causing pollution...". Bu, et al (2026) report device-physics technical details that are tangential to our purposes. Their use of the AHP methodology primarily interests us.

This case study suits the purposes of this investigation for three reasons.

1. *It demonstrates a real-world application.* Introductory AHP presentations often employ contrived cases. A real-world case study provides a more-informative and -convincing demonstration.
2. *This case is recent.* Given that the AHP literature spans decades, a reader might succumb to the temptation to conclude that the method is anachronistic. Our case here demonstrates the application of AHP to a contemporary design problem for an economically essential application.
3. *It strikes a balance between tractability and complexity.* Many pedantic demonstrations are borderline-trivial, constructed to illustrate AHP mechanics. The present investigation benefits from a moderately greater degree of complexity. The selected case provides the opportunity to demonstrate the use of relative entropy to couple hierarchically adjacent hierarchy models.

The following offers neither criticism nor endorsement of Bu, et al (2026). Their work merely serves to illustrate AHP mechanics.

We strive to select AHP terminology that remains close Saaty (1986, 1994). Variations in terminology occur throughout the AHP-application literature. For example, the quantity Saaty (1986) defines as a *priority vector* is frequently referred to as a "weight vector" or "weights". This work remains close to Saaty's conventions.

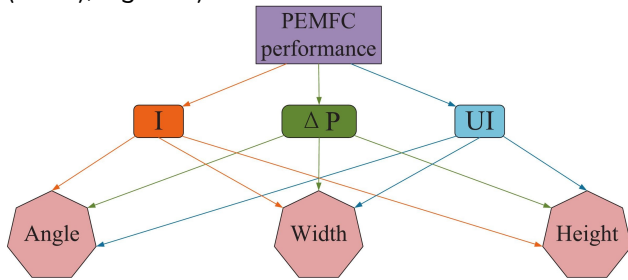
Specification of the evidentiary span.

Figure 2 reproduces the *hierarchical model* employed by Bu, et al (2026) for their fuel-cell design. Visual depiction of the decision-factor hierarchy aids in understanding by human decision-makers. Idiosyncratic representative style notwithstanding, Figure 2 follows the ubiquitous form demonstrated by Saaty (1986, 1994). The elements in Figure 2 map to key DQ links from Figure 1.

- The rectangular box "PEMFC performance" at the top represents the *Goal or Focus*, in Saaty's (1986) terms. A clear goal or focus results from thoughtful framing, in DQ terms.
- The curve-cornered boxes represent *Criteria*. Corresponding to DQ's values, these span range of measurable factors believed to materially distinguish between alternatives. For this case, they represent device physical-operating parameters *current density I* , *pressure drop ΔP* , and *Oxygen-uniformity index UI* . These physical-operating parameters decisively influence the objective.
- The heptagonal shapes at the bottom depict decomposed criteria. Saaty (1986) describes "...intermediate set of higher criteria that can both be decomposed into these attributes and are themselves decompositions of the higher level criteria or subcriteria...". In this case, each of three physical-device design specifications is believed to influence the operating parameters above.

Figure 2 omits the discrete alternatives from which the decision maker selects. Hierarchy-model depictions often include the alternatives when the set is small. Bu, et al (2026) however computationally simulate device performance based on 36 distinct combinations of geometric parameters — angle, width, and height — to construct distinct alternatives. The clutter arising from depicting each scenario in Figure 2 would detract from its clarity. Using AHP to analyze the risk of a large portfolio of distinct assets exemplifies another scenario in which the alternatives do not appear in the hierarchy.

Figure 2: Hierarchy hierarchy used for proton exchange membrane fuel cell (PEMFC) case study. (From Bu, et al (2026), Figure 5)



The criterion hierarchy in Figure 2 is also idiosyncratic for another reason: Each of the physical-specification sub-criteria influences the each operating-parameter criteria. Typically, constituent subcriteria are distinct to their associated criterion. Their representations appear as “clean” tree graph. This excursion from common practice does not invalidate AHPs application, here. One might alternatively consider the Analytic Network Process (ANP) for scenarios like this (Saaty & Vargas (2013); Chen, et al (2019)).

Finally, Figure 2 includes an uncommon feature that clarifies interpretation: It shows edge directivity. Always directed tree graph, representation of AHP hierarchy models tend to omit directional indication. This reinforces our own graphical interpretation of AHP.

Comparative judgements about relative importance of criteria.

AHP imbues values and tradeoffs into decision-making through comparative judgements: “...a matrix to carry out pairwise comparisons of the relative importance of the elements [criteria]...” (emphasis added) (Saaty (1986)). Pairwise is the operative word, here. Practically, one assembles a list of unique, unordered factor pairs for each hierarchy model. If the model for a given objective (or criterion) identifies n elements, then $\binom{n}{2}$ element pairs must be considered.

Figure 2 contains four hierarchic models, each based on three constituent elements. Consequently, $\binom{3}{2} = 3$ element pairs must be considered for each. Each element pair corresponds to a directed edge for the tournament graph by which the hierarchy model is represented.

Table 1 contains the comparative judgements assigned by Bu, et al (2026) for each hierarchy-model element pair depicted in Figure 2. We defer construction of corresponding matrices to the next subsection. Explicit representation here of the comparative judgements contributes clarity of the reasoning. Moreover, the directed-graph edge list in Table 1 provides a key

of departure for subsequent analysis that extends AHP methodology.

The “O-C” model corresponds to the highest level of the Figure 2 hierarchy, relating the relative importance of each of the device operating parameters I , ΔP , and UI . The subsequent hierarchy models specify how Bu, et al (2026) believe that the device-design parameters angle θ , height H , and width W influence the operating parameters.

Now, (Saaty, 1986, Table 1) and (Saaty, 1994, Table 1) specifies a fundamental scale containing subjective heuristics to represent the intensity of importance between criterion pairs. (Bu, et al, 2026, Table 5) adopts this scale, as do other AHP applications appearing in the literature. Saaty (1986), the most-rigorous description of the AHP methodology, contains no explicit justification as to why these specific values are necessary.

Since Saaty (1986) speaks to preferential ordering, alternative, more-rigorous enumerations of importance can be applied. Application of the personal-probability indifference principle occurs as one alternative (Savage (1972); de Finetti (1974)). Hubbard & Seiersen (2023) — addressing the specific domain of cybersecurity — describe more-empirical approaches to measurement. Such techniques could similarly be adapted to AHP comparative judgements.

The construction of Table 1 anticipates its use in defining a directed graph in the sequel. The AHP literature — Saaty (1986, 1994, 2003), in particular — prescribes nothing about the ordering of element pairs in comparative judgement. Construction of Table 1 employed the following convention.

- Populate the “intensity of importance” column with the value greater than unity.
- Assign to the “reference element” column the most-preferred element, associated with the preferred intensity-of-importance value. Subsequently, element corresponds to the head of the directed graph edge connecting the two elements.
- Populate the “comparative element” column the less-preferred element. This element corresponds to the tail of the directed-graph edge.

This convention imbues two conveniences. First, the structure of the table endogenously contains the directive structure of the graph. Second, it inclines us to think of importance intensities in terms as personal probabilities (Savage (1972); de Finetti (1974)) in lieu of the problematic “fundamental scale” from Saaty (1994).

Table 1: Comparative judgements for each hierarchy-model criterion pair. (After Bu, et al (2026), Tables 8, 9, 10, and 11)

hierarchy model	reference element	comparative element	intensity of importance
O-C	I	ΔP	7
O-C	I	UI	4
O-C	UI	ΔP	3
C- ΔP	H	W	7
C- ΔP	H	θ	5
C- ΔP	θ	H	3
C- UI	θ	H	5
C- UI	W	H	5
C- UI	W	θ	3
C- I	W	θ	5
C- I	W	H	3
C- I	H	θ	2

Transformation of comparative judgments into priority vectors.

AHP priority vectors provide the key quantitative mechanism for evidentiary aggregation in MCDM applications. Mechanically, this is accomplished by constructing a reciprocal matrix from the comparative judgements for each hierarchy model. A reciprocal matrix is a Hermitian matrix with unit-valued diagonals and intensity-importance values assigned to off-diagonal elements. For the O-C model in Table 1, this appears as

$$A_{O-C} = \begin{bmatrix} 1 & a_{I,\Delta P} & a_{I,UI} \\ a_{\Delta P,I} & 1 & a_{\Delta P,UI} \\ a_{UI,I} & a_{UI,\Delta P} & 1 \end{bmatrix} \quad (1a)$$

$$= \begin{bmatrix} 1 & 7 & 4 \\ 1/7 & 1 & 1/3 \\ 1/4 & 3 & 1 \end{bmatrix}. \quad (1b)$$

Equation (1a) follows from Table 1 and

$$a_{j,i} = a_{i,j}^{-1},$$

by which we avoid logical contradiction in our reasoning. If

$$A_{O-C} X_{O-C} = X_{O-C} A_{O-C}, \quad (2)$$

represents the eigenvector/eigenvalue decomposition of reciprocal matrix A_{O-C} , then the resulting priority vector ω_{O-C} becomes the Perron (or principal) eigenvector (Saaty (2003)):

$$[\omega_{O-C}]_i = \frac{[X_{O-C}]_{i, \text{argmax}(A_{O-C})}}{\sum_i [X_{O-C}]_{i, \text{argmax}(A_{O-C})}}. \quad (3)$$

Table 2 contains the reciprocal matrix A_{O-C} and the priority vector ω_{O-C} resulting from the comparative judgements for the PEMFC-performance hierarchy model. The results coincide — within roundoff precision — with those reported by Bu, et al (2026).

Table 2: Reciprocal matrix and priority vector resulting from A_{O-C} .

reference factor	I	UI	ΔP	ω_i
I	1	4	7	0.705
UI	$1/4$	1	3	0.211
ΔP	$1/7$	$1/3$	1	0.084

The mechanics illustrated by equations (1a, 2, and 3) are completely general. That ω_{O-C} is a projection onto the unit simplex Δ^3 where

$$\Delta^n = \left\{ x \in \mathbb{R}^n \mid \sum_i x_i = 1 \wedge x_i \geq 0 \forall i \right\}$$

is highly significant. All AHP calculation results are constrained to the interval $[0, 1]$. This includes both AHP-calculation results and standardization of inputs demonstrated in the next subsection.

Entropy of edge-weighted, directed acyclic graph.

Data availability.

This GitHub repository <https://github.com/hamlett-neil-ur/ahp-graph-entropy> contains the data and the python and Neo4j logic on which this paper is based.

References

- Abbas, A. (2026). “Is It Approximate? Or Is It A Fundamentally Flawed Decision-Making Method?” Society of Decision Professionals/INFORMS Decision Analysis Society webinar, May 13, 2026. May 2026 Webinar.
- Brown, D. E., et al (2020). “Entropy of tournament digraph.” *Linear Algebra and its Applications*: 584:371-393. <https://doi.org/10.1016/j.laa.2019.09.026>.
- Bu, W., et al (2026). “Multi-criteria evaluation of symmetric tesla-like valve flow channel geometry on proton exchange membrane fuel cell performance using analytic hierarchy process- entropy weight method.”

- Energy Conversion and Management: 350:120952. <https://doi.org/10.1016/j.enconman.2025.120952>.
- Camuffo, A., Gambardella, A., Pignataro, A. (2024). "Theory-driven strategic-management decisions." *Strategy Science* 9(4):297-514. <https://doi.org/10.1287/stsc.2024.0173>.
- Caporale, D. & Rinaldi, A. (2025). "The application of analytical hierarchy process to assess adaptation strategies for flood and landslides risks: A case study of a multi-risk area community." *Environmental Science & Policy*: 163:103959. <https://doi.org/10.1016/j.envsci.2024.103959>.
- Chen, Z., et al (2015). "Entropy of weighted graph with Randić weights." *Entropy*: 17:3710-3723. <https://doi.org/10.3390/e17063710>.
- Chen, Y., et al (2019).
- Cover, T. M. & Thomas, J. A. (2006). *Elements of Information Theory*. Second edition. New York: Wiley. ISBN: 9780471748823.
- Cox, D. R. & Reid, N. (2000). *The Theory of the Design of Experiments*. UK: Routledge. ISBN: 9781584881957.
- Cox Jr., L. A. (2026a). "Integrating fragmented risk knowledge: Sheaf theory for risk analysts." *Risk Analysis*: 46(3):e70206. <https://doi.org/10.1111/risa.70206>.
- Cox Jr., L. A. (2026b). "Reducing risk misinformation and miscommunication: A sheaf-theoretic perspective." *Risk Analysis*: 46(4):e70224. <https://doi.org/10.1111/risa.70224>.
- Creighton, J. E. (1909). *An Introductory Logic*. London: MacMillan. LCCN: 33034164.
- de Finetti, B. (1974). *Theory of Probability*. Chichester, UK: Wiley. ISBN: 9781119286387.
- Dehmer, M. & Mowshowitz, A. (2011). "A history of graph entropy measures." *Information Sciences*: 181(1):57-78. <https://doi.org/10.1016/j.ins.2010.08.041>.
- Dempster, A. (1967): "Upper and lower probabilities induced by a multivalued mapping." *Annals of Mathematical Statistics*: 38(2):235-339. <https://doi.org/10.1214/aoms/1177698950>.
- Ehrig, T., Felin, T., Zenger, T. (2025). "Causal Reasoning and the Scientific Entrepreneur: Beyond Bayes." *Bayesian Entrepreneurship*. Cambridge, MA: The MIT Press. ISBN: 9780262052160
- Eisenmann, T. (2013, January). "Entrepreneurship: A Working Definition." *Harvard Business Review*. <https://hbr.org/2013/01/what-is-entrepreneurship>.
- Fawcett, T. (2006). "An introduction to ROC analysis." *Pattern Recognition Letters*: 27(8):861-874. <https://doi.org/10.1016/j.patrec.2005.10.010>.
- Guanlei, X., Siogang, X., Xiaotong, W. (2025). "Relative entropy based uncertainty principles for graph signals." *Signal Processing*: 227:109708. <https://doi.org/10.1016/j.sigpro.2024.109708>.
- Hamlett, N. A. (2019). "Competitive advantage based on information." unpublished working paper. <https://drive.google.com/file/d/1Cq7oM6BYrGWxmjf1bdhrdrORZnKBvLu4/view>.
- Hamlett, N. A. (2025). "A theory and methodology for AI-based digital transformation." *Achieving Digital Transformation through Analytics and AI*, J. Liebowitz (ed.). Singapore: World Scientific. https://doi.org/10.1142/9789811296475_0002.
- Hastie, T., Tibshirina, R., Friedman, J. (2009). *The Elements of Statistical Learning*. Springer Nature. <https://doi.org/10.1007/978-0-387-84858-7>.
- Howard, R. A. & Abbas, A. E (2016). *Foundations of Decision Analysis*. New York: Pearson. ISBN: P200000003532.
- Hubbard, D. W. & Seiersen, R. (2023). *How to Measure Anything in CyberSecurity Risk*. New York: Wiley. ISBN: 9781119892335.
- Hu  , S. et al (2026). "Measuring the driving forces of predictive performance: Application to credit scoring." *Management Science*: online first. <https://doi.org/10.1287/mnsc.2023.02025>.
- Jackson, M. C. (2019). *Critical Systems Thinking and the Management of Complexity*. New York: Wiley. ISBN: 9781119118398.
- Jirou  ek, R., Kratochv  l, V., Shenoy, P. P. (2023). "Computing the decomposable entropy of belief-function graphical models." *International Journal of Approximate Reasoning*: 161:108984. <https://doi.org/10.1016/j.ijar.2023.108984>.
- Kay, John & King, Mervyn (2020). *Radical Uncertainty*. New York: Norton. ISBN: 9781324004776.
- Kuo, H. (2022). "Entropy of weighted graph by edge weights by using Sombor reduced index." *Journal of Corrosion and Materials*: 47:20-28. <https://doi.org/10.61336/jcm2022-3>.

- Li, H., et al (2025). "A Bayesian analysis framework for decision making with interval pairwise comparison judgements." *Decision Analysis: "online first"*. <https://doi.org/10.1287/deca.2024.0207>.
- Li, J., et al (2025). "A method for optimizing different geometric shields of D-T neutron generators by combining BP neural network and Analytic Hierarchy Process." *Computer Physics Communications*: 307:109397. <https://doi.org/10.1016/j.cpc.2024.109397>.
- Linkov, I., et al (2006). "Multicriteria decision analysis: A comprehensive decision approach of management of contaminated sediments." *Risk Analysis*: 26(1):61-78. <https://doi.org/10.1111/j.1539-6924.2006.00713.x>.
- Liu, S., et al (2022). "Relative entropy of distance distribution based similarity measure of nodes in weighted graph data." *Entropy*: 24:1154. <https://doi.org/10.3390/e24081154>.
- Lukinskiy, Valery, Lukinskly, Vladislav, Bazhina, D. (2025). "Refining the analytic hierarchy process: A statistical and methodological exploration of expert judgments." *Alexandria Engineering Journal*: 125:526-536. <https://doi.org/10.1016/j.aej.2025.04.022>.
- Martin, R. L., & Osberg, S. (2007). *Social entrepreneurship: The case for definition*. Stanford Social Innovation Review, 2007 (Spring). <https://www.ngobiz.org/picture/File/Social%20Entrepreneur-The%20Case%20of%20Definition.pdf>.
- Merrick, J. R. W., van Dorp, J. R., Singh, A. (2005). "Analysis of correlated expert judgements from extended pairwise comparisons." *Decision Analysis*: 2(1):1-65. <https://doi.org/10.1287/deca.1050.0031>.
- Moon, J. W. (2015). *Topics on Tournaments in graph Theory*. Dover. ISBN: 9780486804736. (Originally, New York: Holt, Rinehart, and Winston 1968).
- Mowshowitz, A. & Dehmer, M. (2012). "Entropy and the complexity of graph revisited." *Entropy*: 14:559-570. <https://doi.org/10.3390/e14030559>.
- Olive, D. J. (2017). *Linear Regression*. Cham, CH: Springer Nature. <https://doi.org/10.1007/978-3-319-55252-1>.
- Pearl, J. (1988). *Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference*. Elsevier. ISBN: 9780080514895.
- Rindova, V. P. & Martins, L. L. (2024). "The imagination advantage: Why and how strategists combine knowledge and imagination in developing theories." *Strategy Science*: 9(4): 297-314. <https://doi.org/10.1287/stsc.2024.0184>.
- Rosiak, D. (2022). *Sheaf Theory through Examples*. Cambridge, MA: The MIT Press. ISBN: 9780262362375. <https://doi.org/10.1287/inte.24.6.19>.
- Saaty, T. L. (1986). "Axiomatic foundation of the Analytic Hierarchy Process." *Management Science*: 32(7):841-855. <https://doi.org/10.1287/mnsc.32.7.841>.
- Saaty, T. L. (1987). "Risk — Its priority and probability: The Analytic Hierarchy Process." *Risk Analysis*: 7(2):159-172. <https://doi.org/10.1111/j.1539-6924.1987.tb00980.x>.
- Saaty, T. L. (1994). "How to make a decision: The Analytic Hierarchy Process." *Applied Analytics*: (24)6:19-43. <https://doi.org/10.1287/inte.24.6.19>.
- Saaty, T. L. (2003). "Decision-making with the AHP: Why is the principal eigenvector necessary." *European Journal of Operational Research*: 145(1):85-91. [https://doi.org/10.1016/S0377-2217\(02\)00227-8](https://doi.org/10.1016/S0377-2217(02)00227-8).
- Saaty, T. L. & Vargas, L. G. (2013). *Decision Making with the Analytic Network Process*. New York: Springer. ISBN: 9781461472797.
- Sáenz-Royo, C. & Chiclana, F. (2026). "Contribution of the ratio scale of expert judgments in the analytic hierarchy process." *Expert Systems with Applications*: 296(C):128938. <https://doi.org/10.1016/j.eswa.2025.128938>.
- Saliconi, S. & Prioli, M (2018). *Measuring Uncertainty within the Theory of Evidence*. Cham, CH: Springer Nature. <https://doi.org/10.1007/978-3-319-74139-0>.
- Savage, L. (1972). *The Foundations of Statistics*. Garden City, NY: Dover. ISBN: 9780486137100. (Originally, Wiley, 1954.)
- Shafer, G. (1976). *A Mathematical Theory of Evidence*. Princeton, NJ: Princeton University Press. ISBN: 9780691100425.
- Shenoy, P. P. (2023). "Making inferences in incomplete Bayesian networks: A Dempster-Shafer belief-function approach." *International Journal of Approximate Reasoning*: 160(2023):108967. <https://doi.org/10.1016/j.ijar.2023.108967>.
- Spetzler, C., Winter, H., Meyer, J. (2016). *Decision Quality Value Creation from Better Business Decisions*. New York: Wiley. ISBN: 9781119176657.

- Swanson, E. B. (2024). "Technology entrepreneurship is more than one might think." *Information and Organization*: 34:100512. <https://doi.org/10.1016/j.infoandorg.2024.100512>.
- Townsend, D. M., et al (2023). "Are futures computable? Knightian uncertainty and artificial intelligence." *Academy of Management Review*. (50)2:415:440. <https://doi.org/10.5465/amr.2022.0237>.
- Vaidya, O. S. & Kumar S. (2006). "Analytical hierarchy process: An overview of applications." *European Journal of Operational Research*: 169(1):1-29. <https://doi.org/10.1016/j.ejor.2004.04.028>.
- von Hayek, Fredrick (1974), "The pretence of knowledge", Nobel Prize Lecture, December 11, 1974, <https://www.nobelprize.org/prizes/economic-sciences/1974/hayek/lecture/>.
- Wong, A. K. C. & You, M. (1985). "Entropy and distance of random graph with application to structural pattern recognition." *IEEE Transactions on Pattern Analysis and Machine Intelligence*: PAMI-7(5):599-609. <https://doi.org/10.1109/TPAMI.1985.4767707>.
- Xiao, H., et al (2025). "A simulation optimization approach for weight valuation in analytic hierarchy process." *European Journal of Operations Research*: 321(3):851-864. <https://doi.org/10.1016/j.ejor.2024.10.018>.
- Zellweger, T., & Zenger, T. (2023). "Entrepreneurs as scientists: A pragmatist's approach to producing value out of uncertainty." *Academy of Management Review*: 48(3):397-408. <https://doi.org/10.5465/amr.2020.0503>.
- Zeng, D., et al (2025). "Material selection of titanium alloy pipelines considering multi-criteria in an acidic environment based on analytic hierarchy process." *International Journal of Pressure Vessels and Piping*: 2016:105461. <https://doi.org/10.1016/j.ijpvp.2025.105461>.